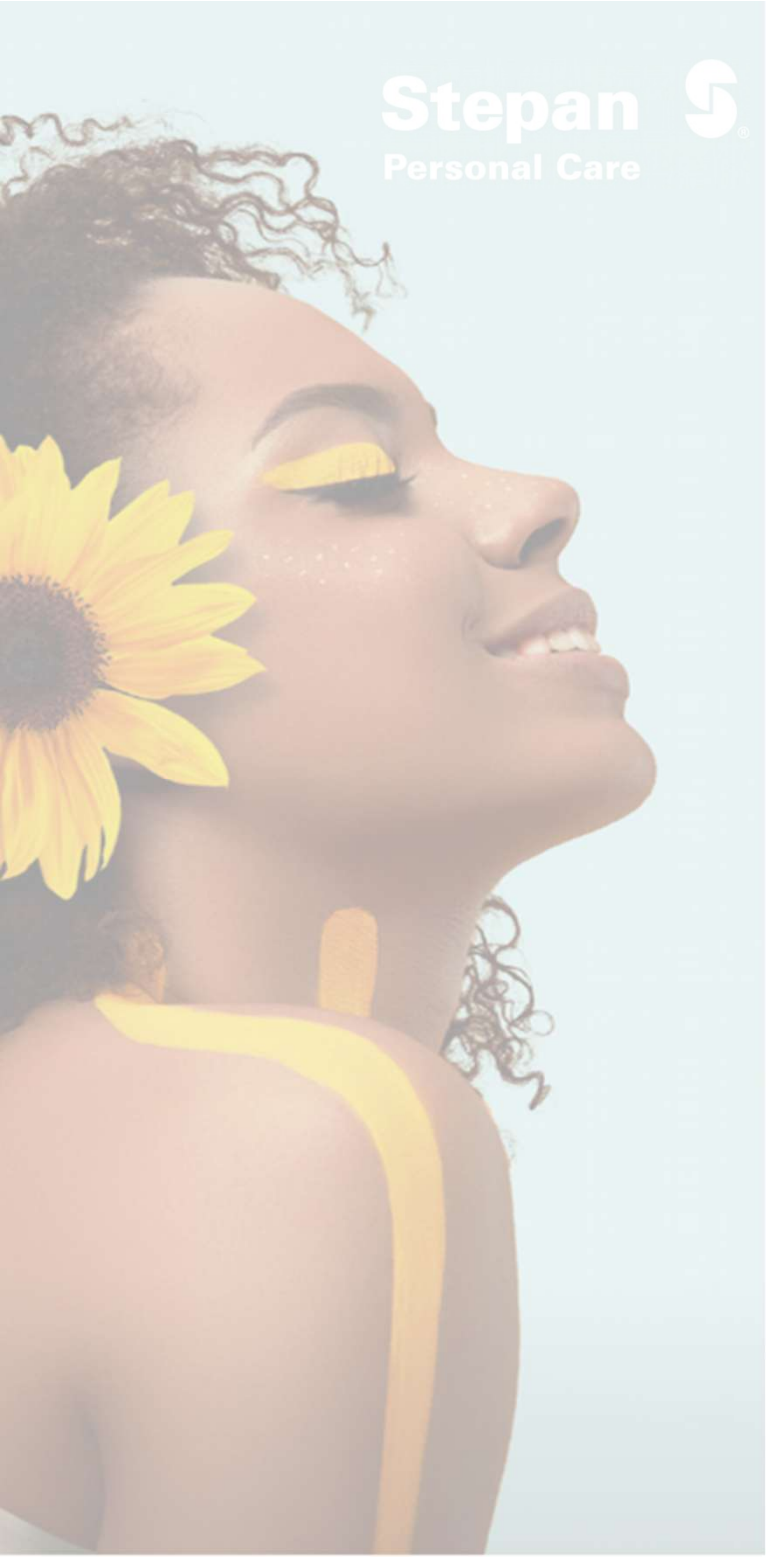


Sustainably Sourced Ingredients for Haircare

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Page withheld in this copy: it held internal reviewer comments, not seminar content.

Do not edit
How to change the design



What are the main constraints you experience when formulating a hair conditioner?

The Slido app must be installed on every computer you're presenting from

slido

Sustainability in Beauty

Adapting to greener consumers



>50%

Of consumers trust labels with locally sourced, environmentally friendly and natural[2]

68% of consumers stated that doing things that benefit the environment makes me feel happy[3]

27% of consumers seek natural ingredients in their regular hair care products[1]



Companies planning to develop and launch product with sustainability attributes[1]

58%

Sustainable sourcing claims

70%

Environmentally friendly claims

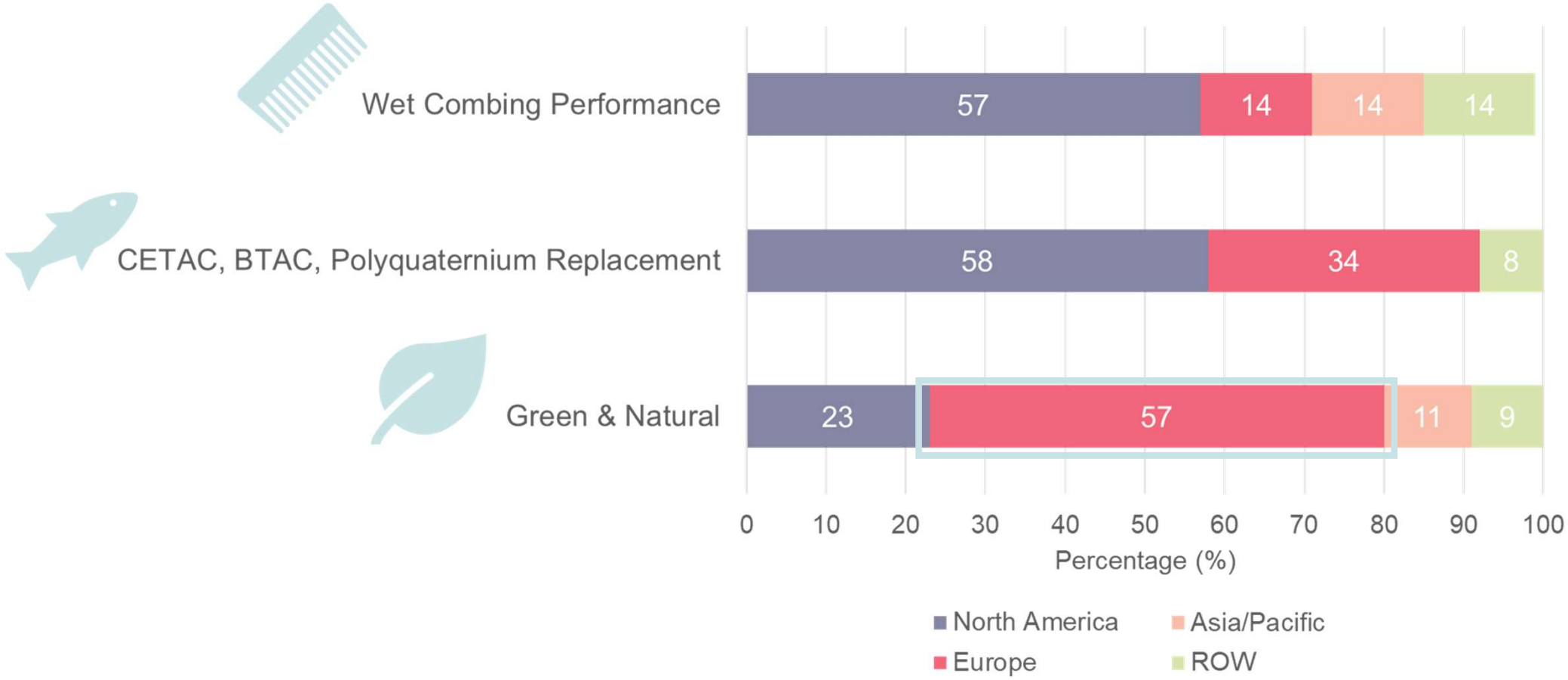




Beauty consumers want to be closer to the manufacturing process than just receiving the final product, they are curious where products have been produced and how beauty brands are contributing to a more sustainable environment[3]

1 Source: Euromonitor 2022
2 Source: Euromonitor 2019
3 Source: Mintel 2022

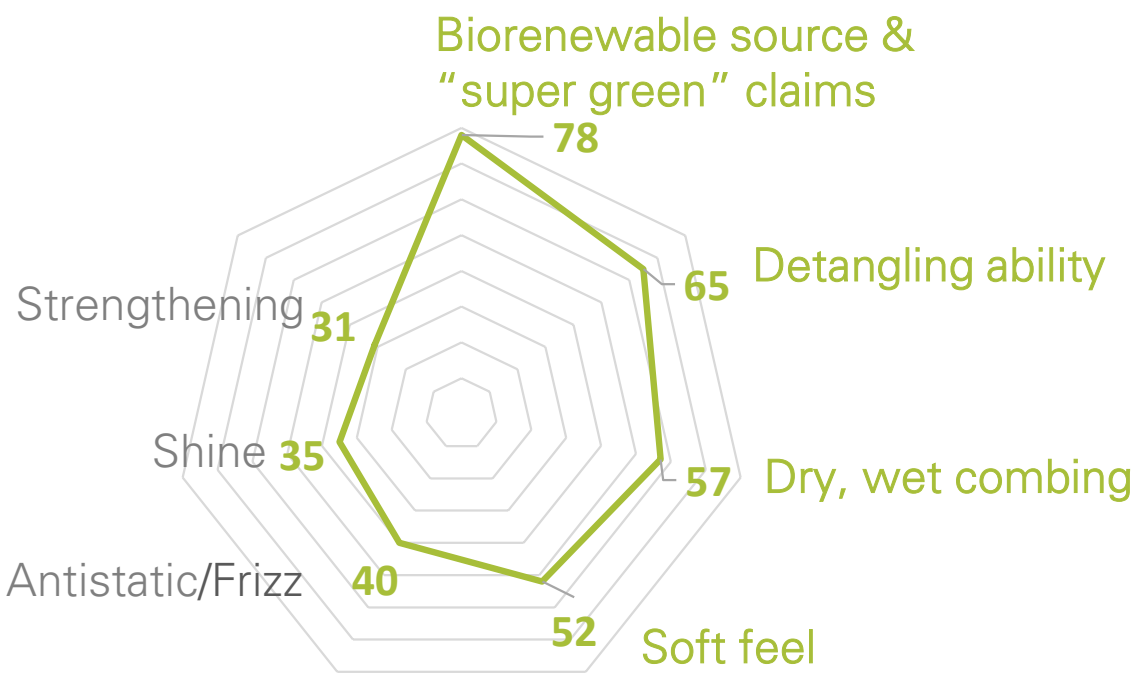
Sustainability for Brands



Stepan Market Research Study, 2021
BTAC = Behentrimonium Chloride
CETAC = Cetrimonium Chloride
ROW = Rest of world

Sustainability for Formulators

What are the critical needs for your formulation?



Biorenewable source claims top interest

Detangling ability noted as 2nd critical need, even before dry and wet combability

The need for soft feel also noted by >50% of respondents

Stepan Market Research Study, 2021

Sustainability for Formulators

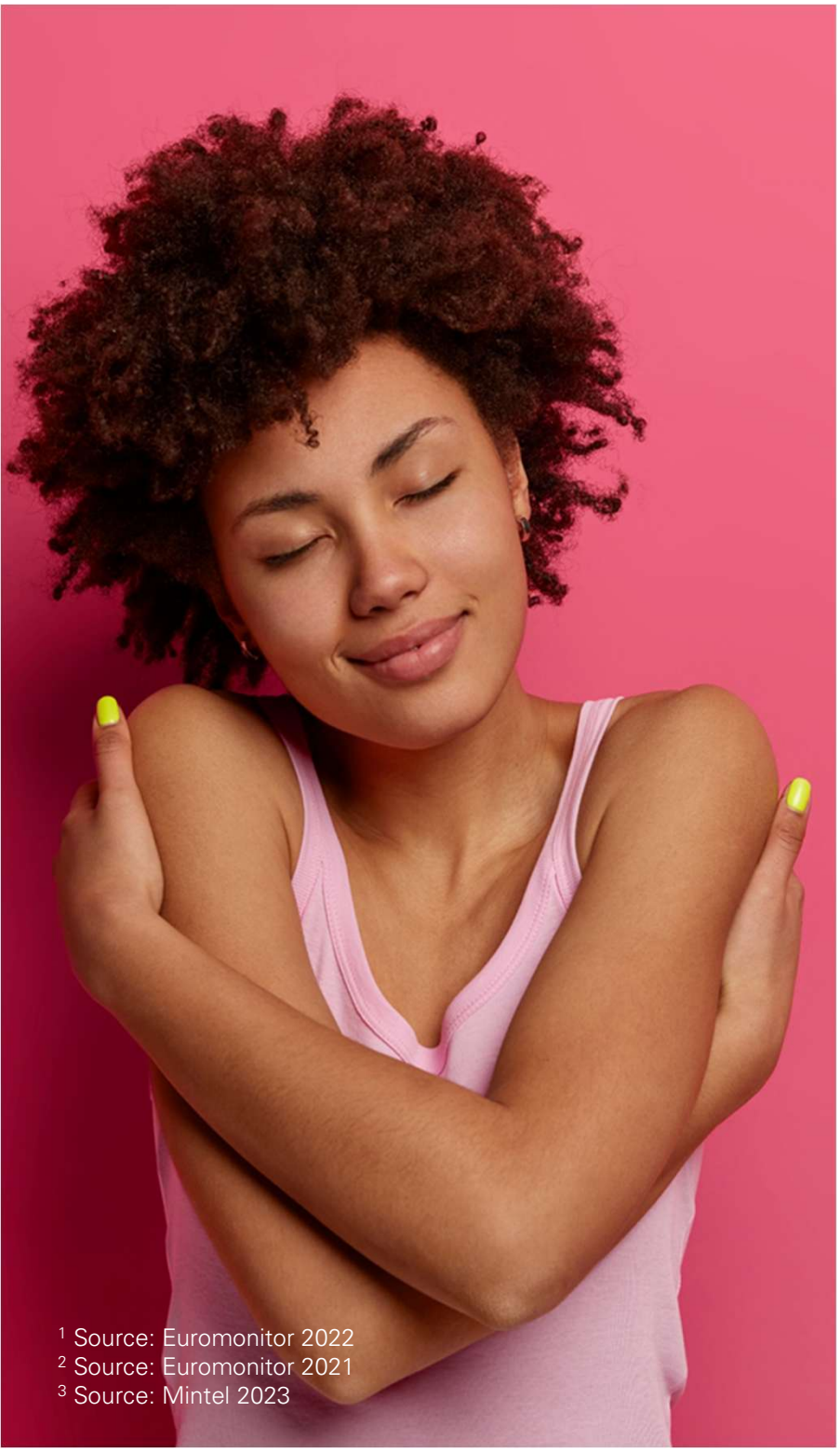
The top 3 issues formulators face when developing sustainable products

Lack of smooth and soft feel compared to conventional ingredients

Need to use higher percentage for similar performance, increasing formulation cost

Often difficult to formulate

Stepan Market Research Study, 2021



The Essential You in Hair Care

"I am who I am"
Breaking stereotypes and societal norms, allowing Self-love Seekers to express their true selves[1]

Empathy
Acceptance, self care and inclusion are at the forefront of consumer lifestyle[1]

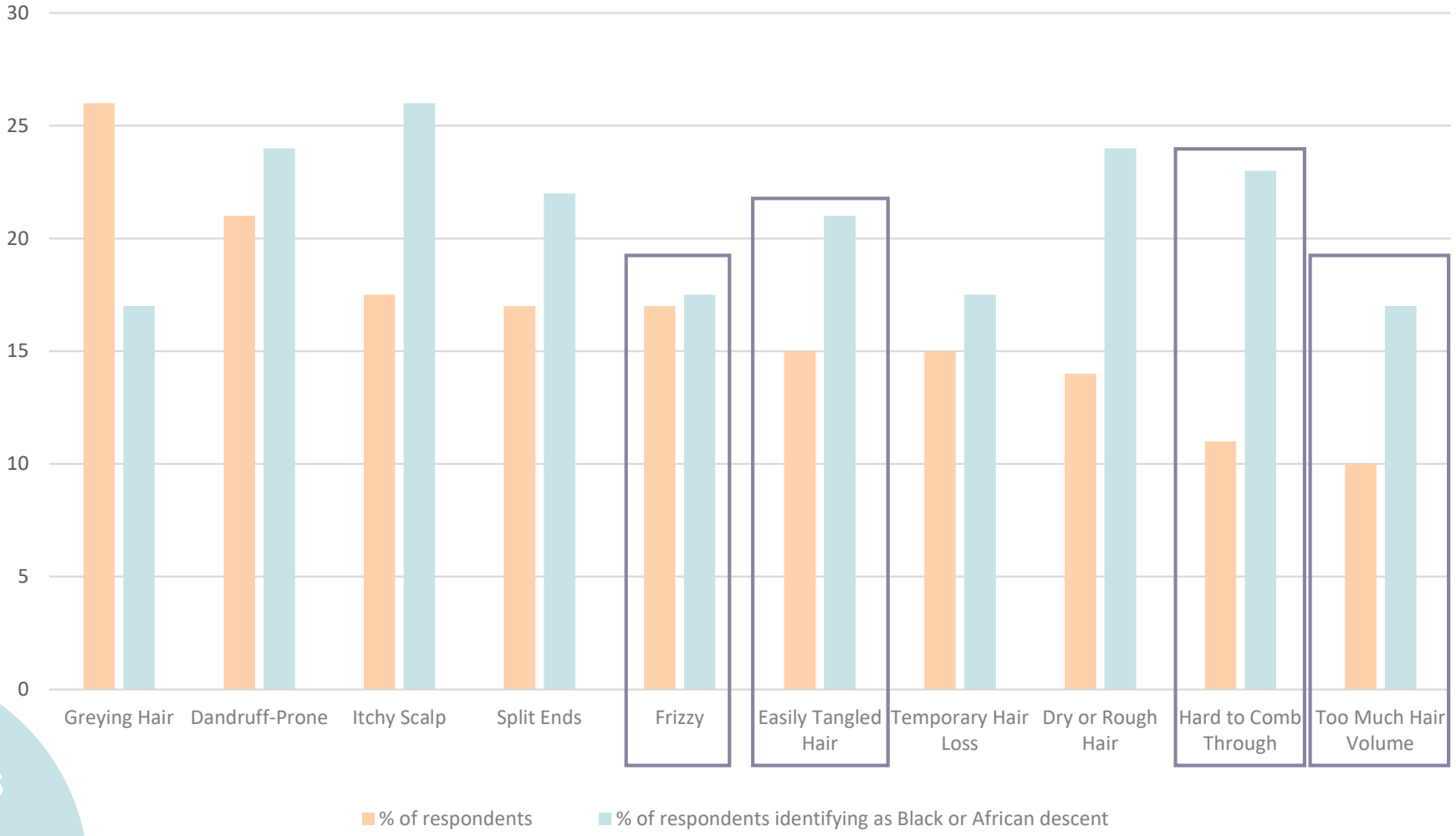
Me Mentality
Consumers will be eager to re-focus on themselves[3]

Identifying Needs
Ingredient companies are looking to expand their expertise in "textured hair"[2]

[1] Source: Euromonitor 2022
[2] Source: Euromonitor 2021
[3] Source: Mintel 2023

Come and learn more at Stepan Stand AE80

Top 10 Reported Hair Concerns

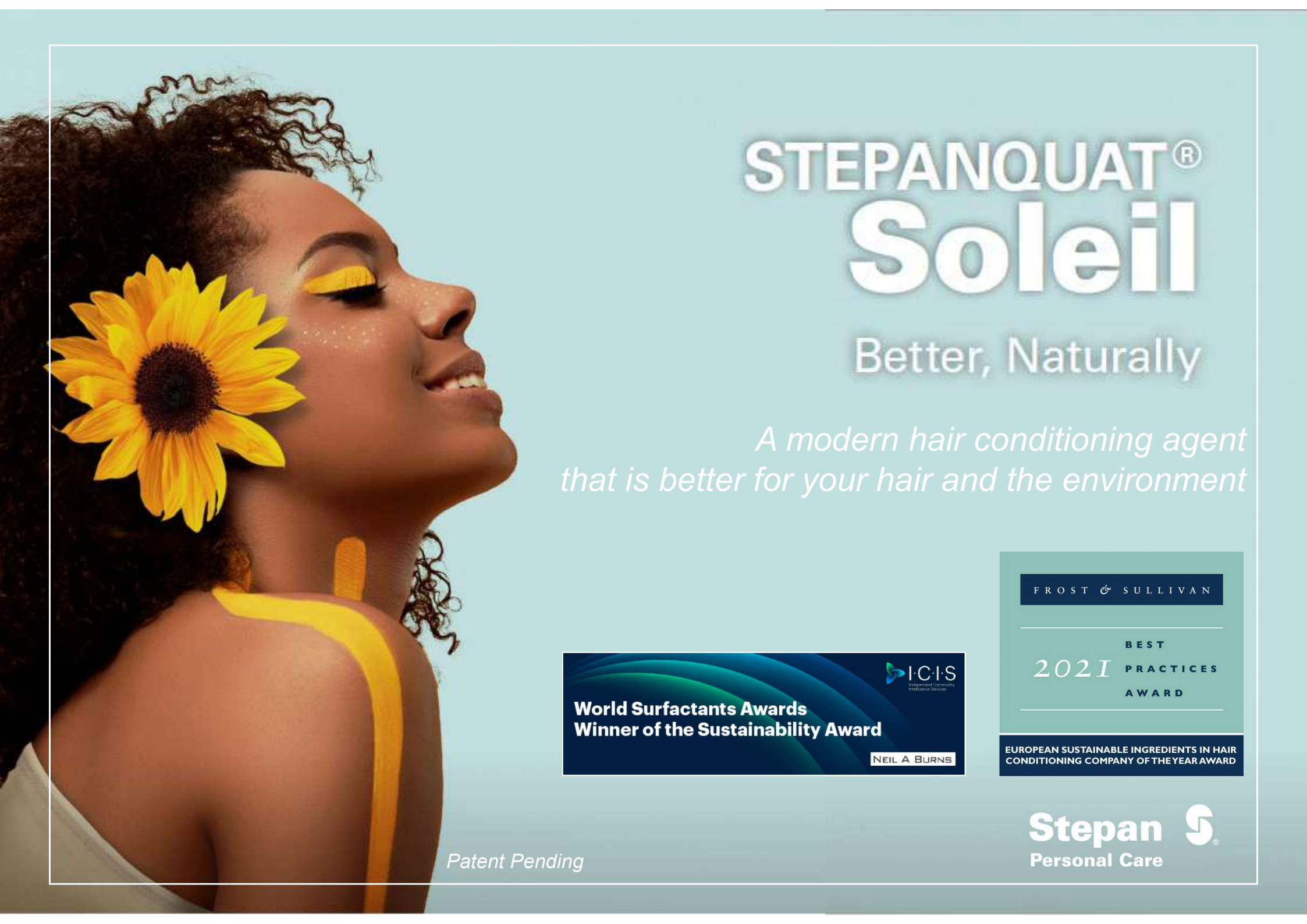


Top Benefits

Frizz Fighting
Volumizing

Stepan Market Survey 2022

Source: Euromonitor International's Beauty Survey, 2021 Source


STEPANQUAT®
Soleil
Better, Naturally
A modern hair conditioning agent
that is better for your hair and the environment
I·C·I·S
World Surfactants Awards
Winner of the Sustainability Award
NEIL A BURNS
FROST & SULLIVAN
2021 BEST PRACTICES AWARD
EUROPEAN SUSTAINABLE INGREDIENTS IN HAIR CONDITIONING COMPANY OF THE YEAR AWARD
Patent Pending
Stepan
Personal Care

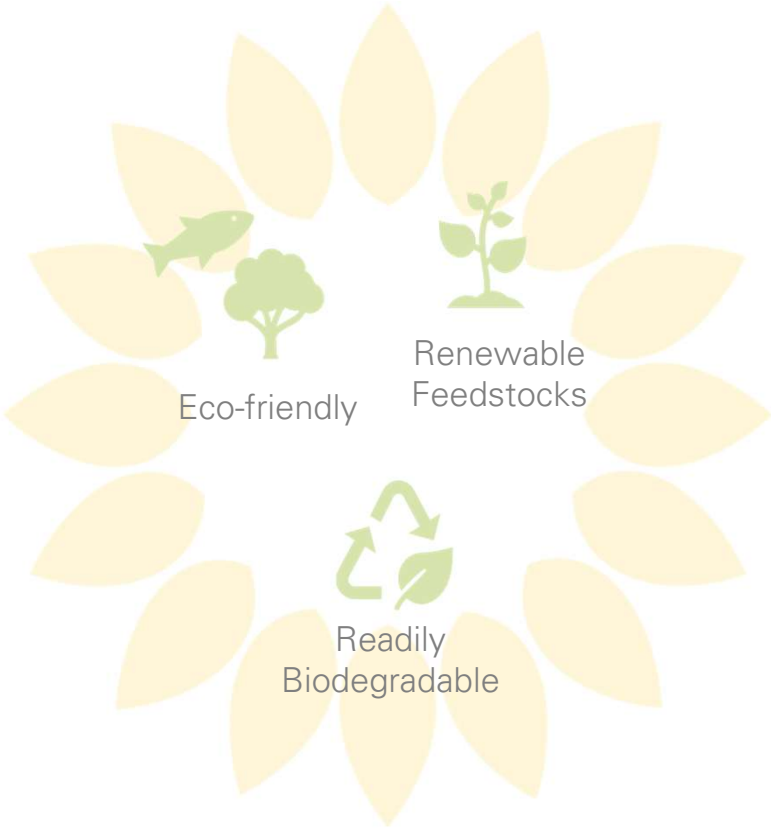
Next Generation Conditioning Agent

Better, Naturally



STEPANQUAT® Soleil

Ingredients	Dioleoylethyl Hydroxyethylmonium Methosulfate (and) Sunflower Seed Oil Glycerides
Source	Non-GMO, EU-Grown Sunflower Oil
ISO 16128 Score*	0.88
Actives %	100
Free From	Silicones, Preservatives, Glycols & Ethanol
Registrations	Global Compliance



*Stepan calculates Natural Origin Content based on biorenewable carbon content and excludes balance water and impurities/byproducts

Easy-to-use for Formulators

Addressing the 1st and 2nd top issues formulators face with current sustainable solutions

Easy-to-Use 100% active liquid

No Regulatory Use-Level Restrictions

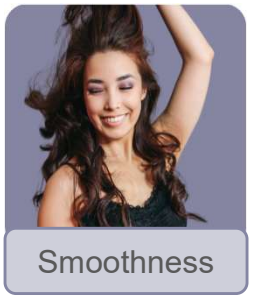
No Special Storage or Handling Requirements

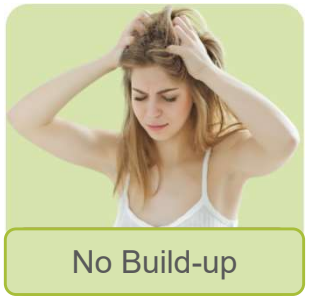


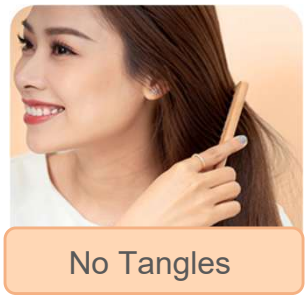
Versatile for use in:

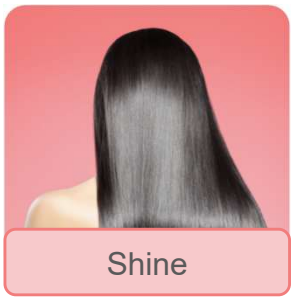
- Conditioners
- Treatments
- Masks
- Deep Conditioners
- Leave-in Products
- Styling Products
- Shampoo

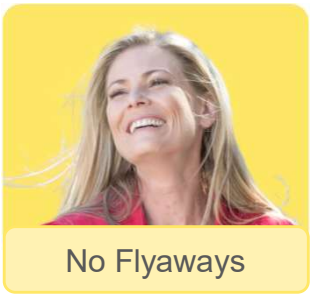
Claim Substantiation

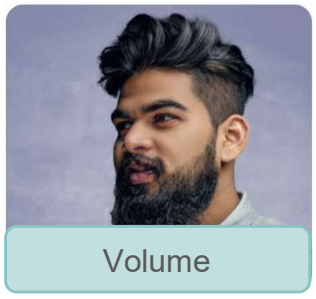

Smoothness


No Build-up


No Tangles


Shine


No Flyaways


Volume

	Smoothness	No Build-up	No Tangles	Shine	No Flyaways	Volume
Instrumentation	Scanning electron microscopy	Rubine dye test	Dia-Stron MTT690	BossaNova SAMBA	BossaNova BOLERO	BossaNova BOLERO
Claim	Smoothness	No build-up	Dry combing, wet combing	Shine	Flyaway reduction	Volume
Output	Hair structure imaging	Dye does not stick to the hair	Force value to comb	Luster value image analysis	% change value image analysis	% change value image analysis

Standardized formula, with adjusted final pH of of 3.5-4.5

Constant Concentration

Conventional Benchmarks
- BTAC
- CETAC

Hair Conditioner Formula*

Ingredient	*% by weight*
Deionized Water	q.s. to 100.00
Hydroxyethylcelullose	0.70
Conditioning Active	**2% active**
Cetyl Alcohol	2.00
Potassium Chloride	0.50



*Common chassis used for all claims substantiation, except for smoothness, curl control where chassis does not contain any hydroxythylcellulose and conditioning active is reduced to 1.35%

Embracing Myself and My Hair Type

My Value, My Identity

Express My Individuality

Andre Walker's Hair Typing Scale

Type One	Straight Hair Fine & fragile to coarse & thin	A B C
Type Two	Wavy Hair Fine & thin to coarse & frizzy	A B C
Type Three	Curly Hair Lose to corkscrew curls	A B
Type Four	Kinky Hair Tight to Z-angled coils	A B

Straight



Wavy



Curly



Kinky



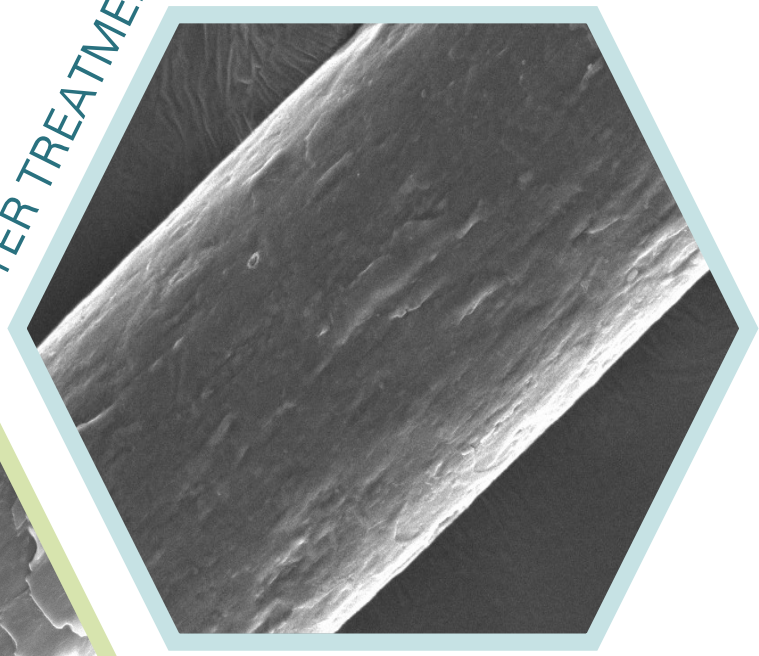


Smoothness

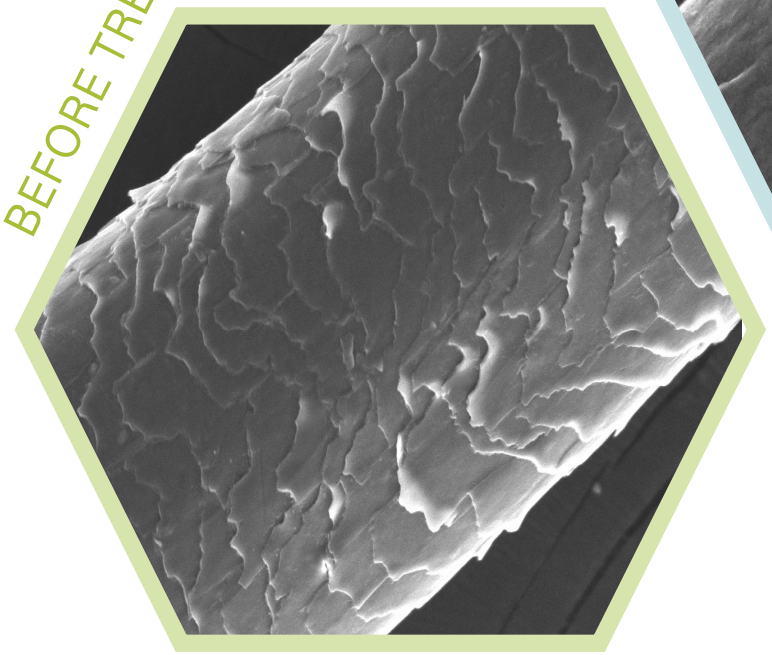
STEPANQUAT Soleil smooths each individual hair strand for super-soft, manageable hair

STEPANQUAT Soleil

AFTER TREATMENT

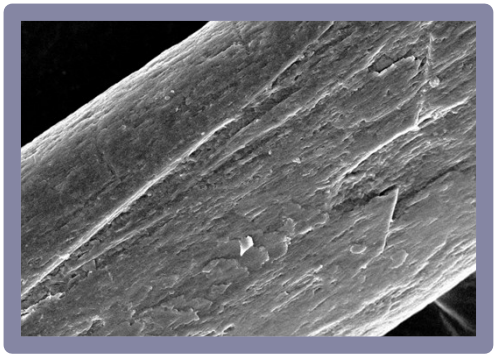


BEFORE TREATMENT

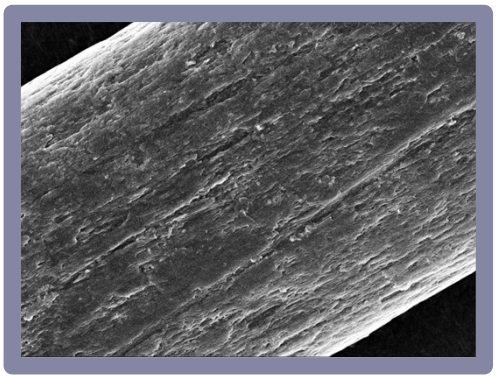


CETAC

AFTER TREATMENT



BTAC



CETAC = cetrimonium chloride
BTAC = benhentrimonium chloride

Scanning Electron Microscopy images of Caucasian, slightly bleached hair tresses treated with a 2% active conditioner at 1,000x magnification



Smoothness

STEPANQUAT Soleil to the rescue of unruly hair



2B type wavy hair washed with a standard shampoo then treated with a rinse-off conditioner at 1.35% active

Hair smoothing performance for curl control



Hair Conditioner Formula

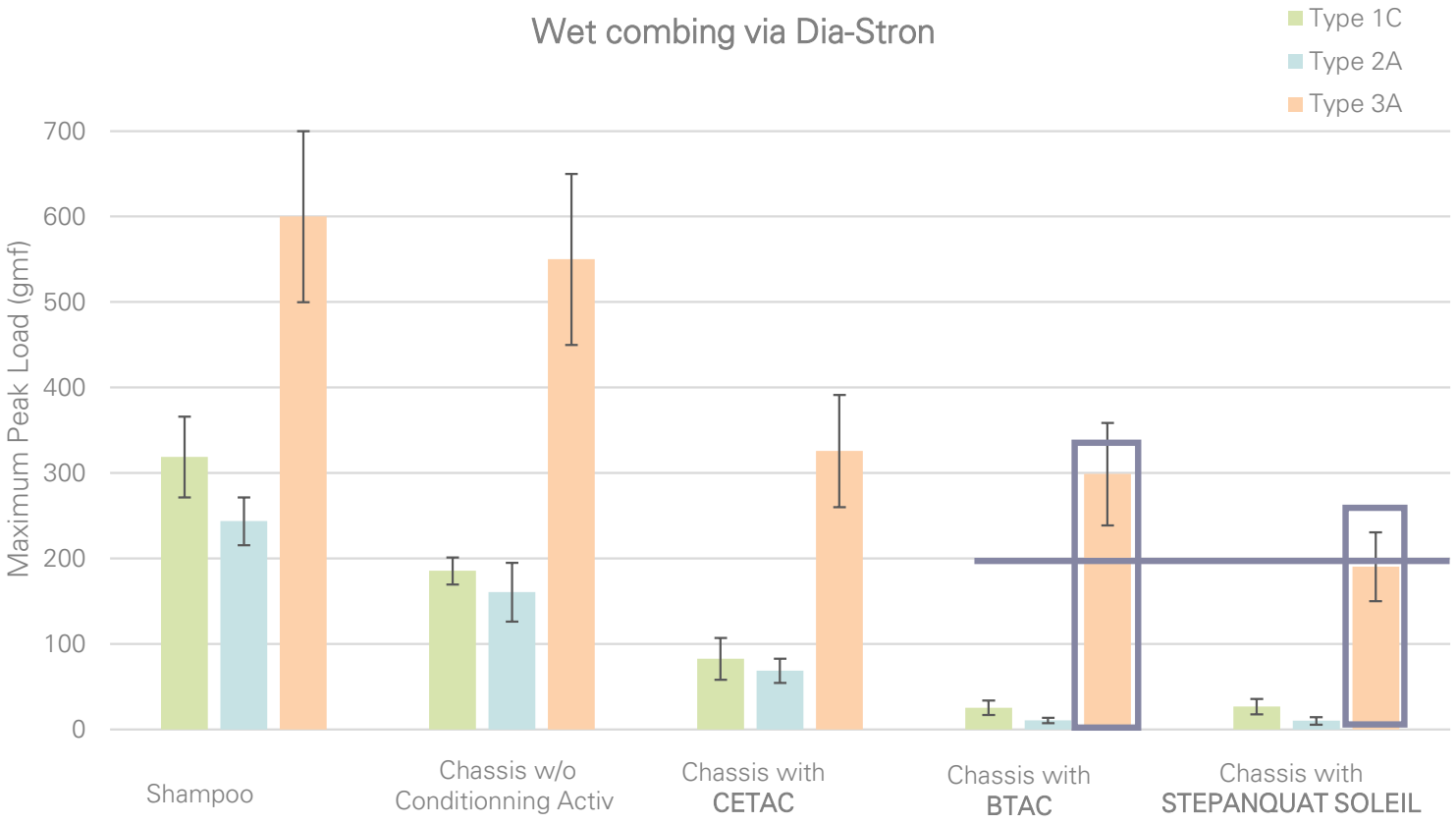
Ingredient	% by weight
Deionized Water	q.s. to 100.00
Conditioning Active	1.35% active
Cetearyl Alcohol	3.35
Preservative	q.S
Sequestrant	q.s



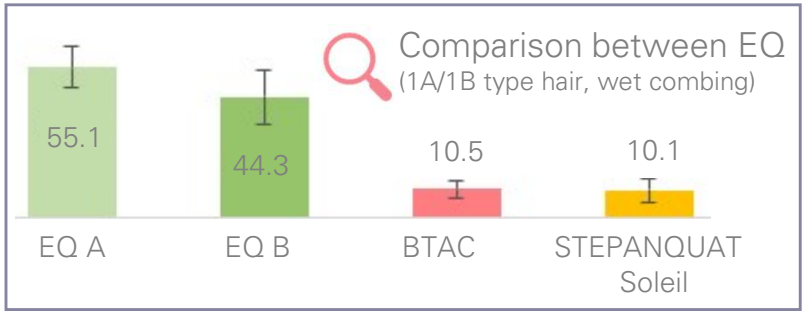
No Tangles

STEPANQUAT Soleil is equal to/or outperforms BTAC

Decreased Tangles







Testing parameters = three tresses with 10 comb-throughs each

CETAC = Cetrimonium Chloride
BTAC = Behentrimonium Chloride
EQ A = Distearoylethyl Dimonium Chloride
EQ B = Dipalmitoylethyl Hydroxyethyl Ammonium Methosulfate



No Build up

STEPANQUAT Soleil possesses great substantivity

Modified rubine dye test: detection of cationics on keratin thanks to anionic dye deposit
Images of Caucasian, slightly-bleached hair tresses treated with a 2% active conditioner

HAIR BEAUTY ROUTINE!

Let's get a shampoo!
With or w/o condition?

Now, let's get a shampoo again!

NO TREATMENT



Blond uniform

STEPS

1. Let's get a shampoo!
2. Rinse
3. With or without conditioning agent?
4. Rubine red dye application
5. Rinse
6. Dry

WITHOUT conditioning agent after step 6

CONTROL



WITH conditioning agent after step 6

BTAC



Pale not uniform

Blond uniform hair tresses

STEPANQUAT SOLEIL



Deep red uniform

STEPS

7. Now let's get a shampoo again
8. Rinse
9. Rubine red dye re-application
10. Rinse
11. Dry

The deeper the red color
The better the deposit of the red dye
The better the substantivity!

STEPANQUAT Soleil
Easily washes up
No build up effect



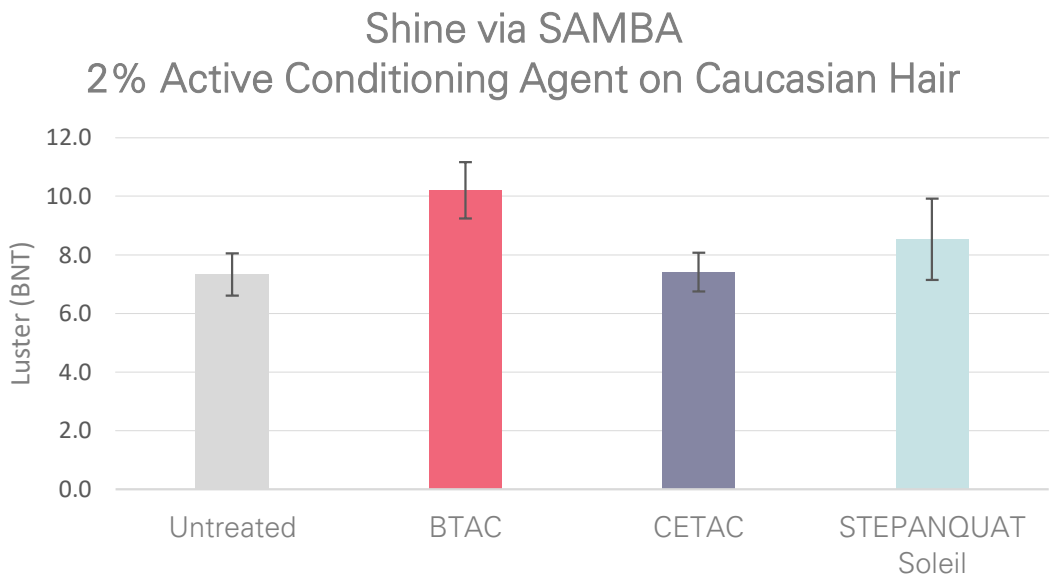
Blond uniform



Shine

STEPANQUAT Soleil performs equally to BTAC

Increased Shine





Testing parameters = six tresses with 10 measurements each

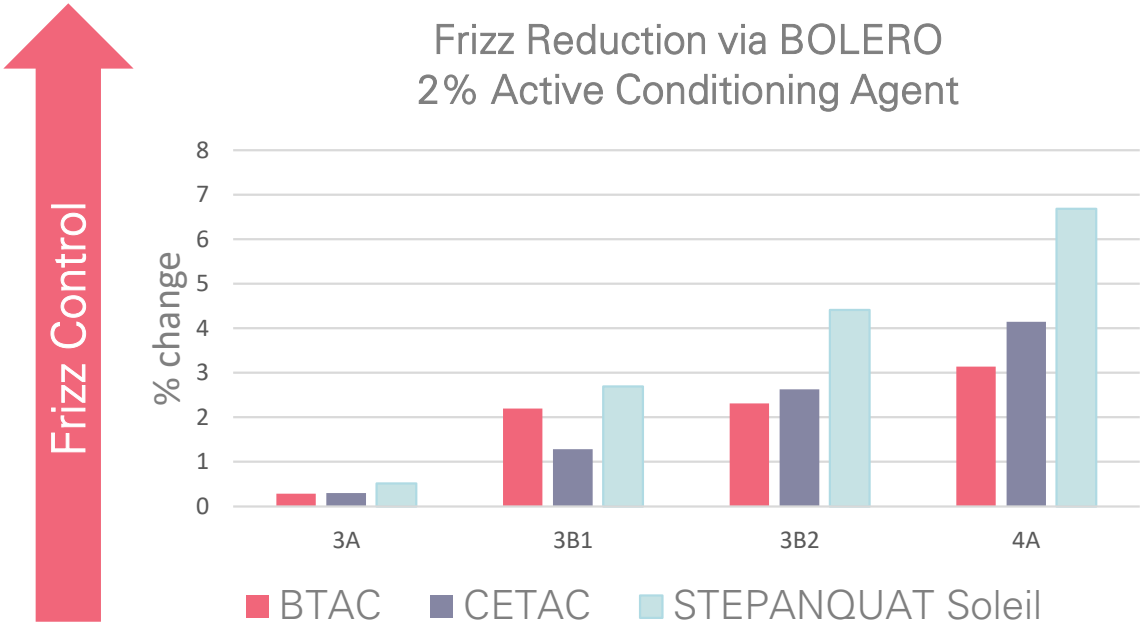


No Flyaways

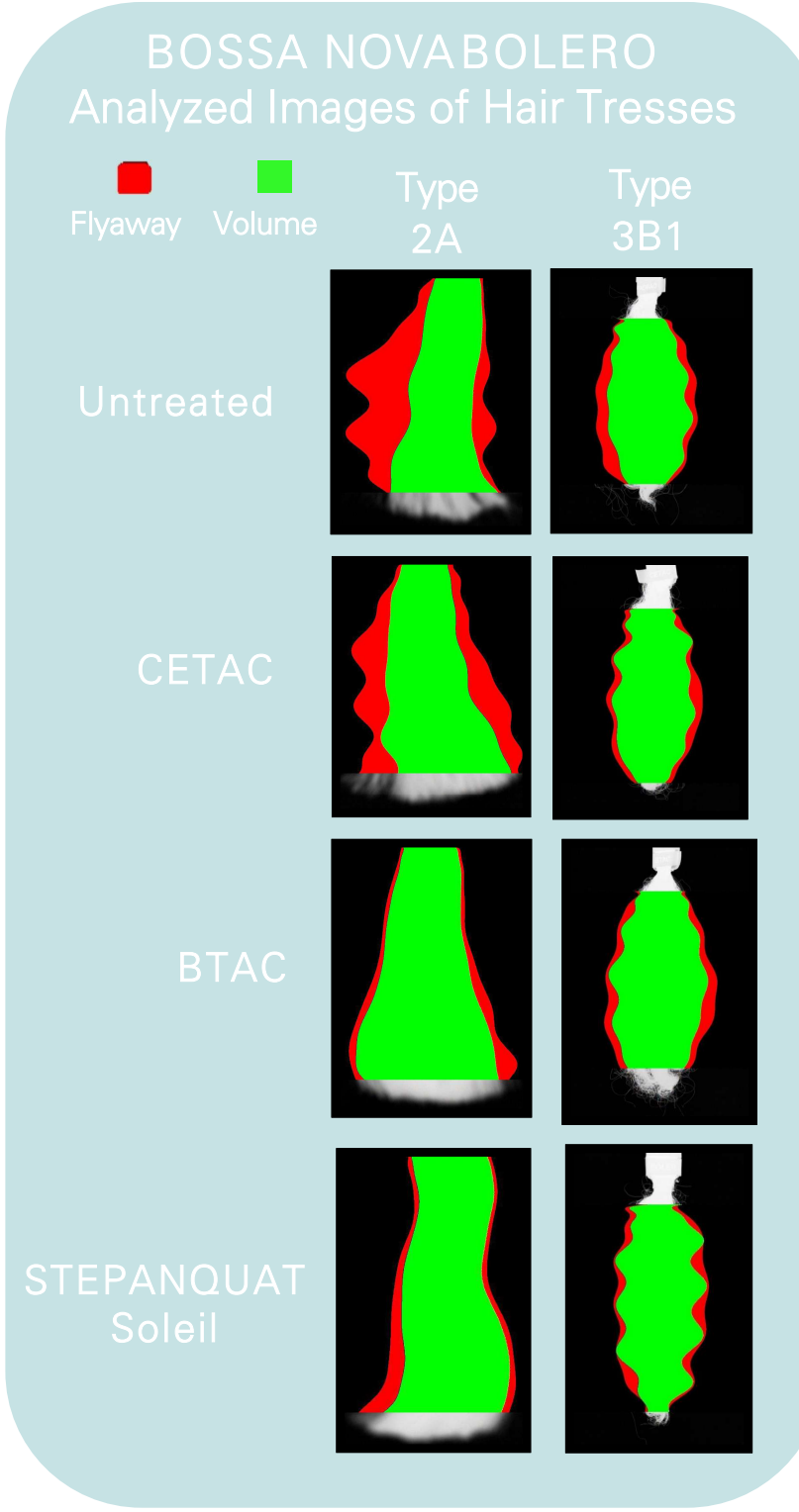
STEPANQUAT Soleil outperforms CETAC

2A Hair Type: STEPANQUAT Soleil outperforms CETAC and performs equal to BTAC for frizz reduction or for flyaway reduction and increased volume (see images)

Textured Hair (Types 3 and 4): STEPANQUAT Soleil performs equal to or reduces frizz slightly more than leading conditioning agents



Testing parameters = 6 tresses with 10 measurements each

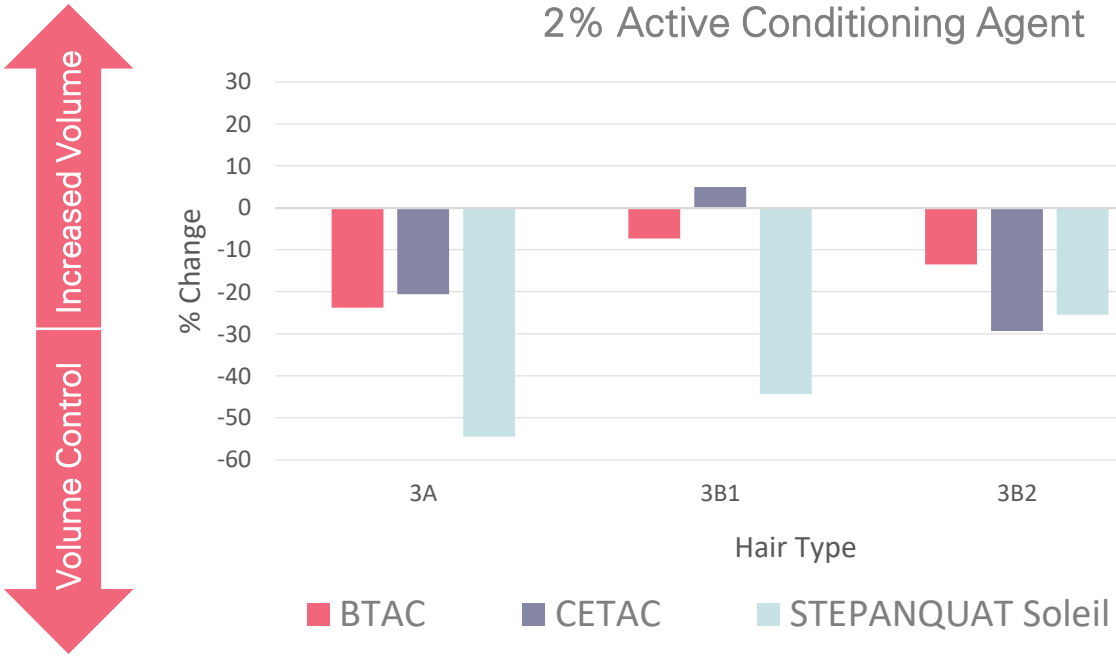




Volume

STEPANQUAT Soleil excels for textured hair care

STEPANQUAT Soleil controls hair volume as well as or better than leading conditioning agents, resulting in tamer curls and a sleeker look.



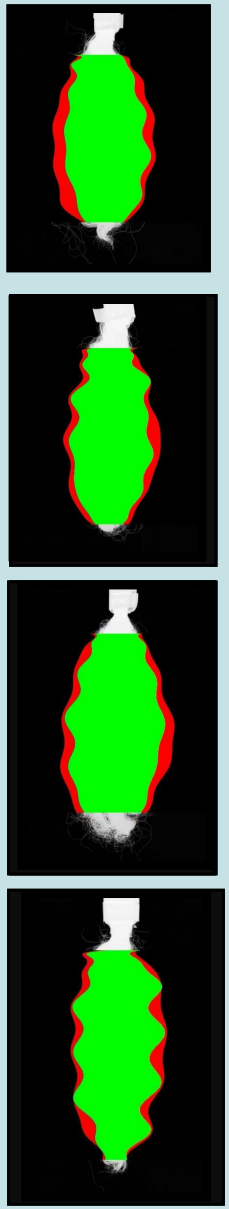


Testing parameters = 6 tresses with 10 measurements each

BOSSA NOVA BOLERO
Analyzed Images of Hair tresses

Flyaway Volume

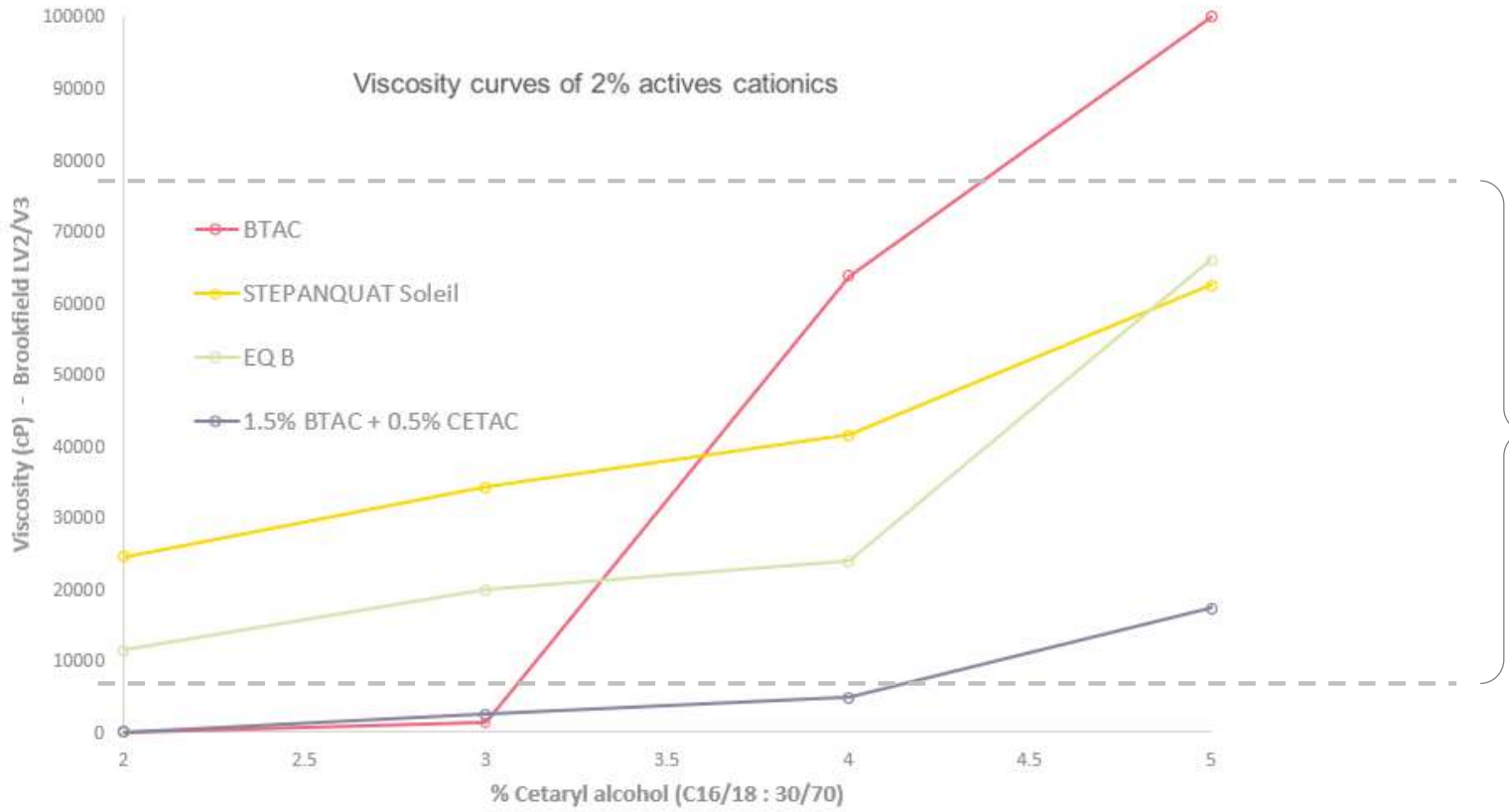
Type 3B1



Formulation Guidelines

STEPANQUAT Soleil offers a regular and smooth viscosity behaviour

STEPANQUAT Soleil brings viscosity with lower amounts of fatty alcohol compared to other cationics and offers a gradual thickening pattern



Viscosity range of benchmark hair conditioners & masks

STEPANQUAT Soleil provides the target viscosity for conditioners and masks at 2-5% cetearyl alcohol.

CETAC = Cetrimonium Chloride
BTAC = Behentrimonium Chloride
EQ A = Distearoylethyl Dimonium Chloride
EQ B = Dipalmitoylethyl Hydroxyethyl Ammonium Methosulfate

Formulation Guidelines

Making a Conditioner

Phase	Ingredients	w/w %
A	Aqua/Water	Up to 100
	Euxyl K712 – Potassium Sorbate & Sodium Benzoate (Schülke & Mayr SA)	0.4
B	Cetearyl Alcohol	5.0
	STEPANQUAT Soleil – Dioleoylethyl Hydroxyethylmonium Methosulfate & Sunflower Seed Oil Glycerides	2.0
	Helianthus Anuus (Sunflower) Seed Oil	1.2
C	Tocopherol	0.2
	Euxyl K712 – Sodium Benzoate & Potassium Sorbate (Schülke & Mayr SA)	0.4
	Fragrance – Silky Touch (Mane & Fils SA)	0.2



Phase A: Warm water until 65°C then add Euxyl K712.
Phase B: Melt ingredients together at 65°C, then add to Phase A.
Use a dispersion mix to emulsify.
Slowly cool down the emulsion until 40°C.
Phase C: Introduce ingredients in given order.
Control and adjust pH, if necessary, with citric acid.

Controls: pH = 2.9 | Viscosity (Brookfield, LV2 V3, at 20°C) = 32100 cP

Euxyl K712 needs to be added at two different times to maintain the viscosity.
Alternative preservatives for a single addition: Benzyl Alcohol & Ethylhexyl Glycerine or Phenoxyethanol

Formulation Guidelines

Making a Hair Mask



Phase	Ingredients	w/w %
A	Aqua/Water	Up to 100
B	Cetearyl Alcohol	5.0
	STEPANQUAT Soleil – Dioleoylethyl Hydroxyethylmonium Methosulfate & Sunflower Seed Oil Glycerides	2.0
	Butyrospermum Parkii (Shea) butter	0.9
C	Tocopherol	0.2
	Euxyl K900 – Benzyl Alcohol & Ethylhexyl Glycerine (Schülke & Mayr SA)	0.8
	Fragrance – Silky Touch (Mane & Fils SA)	0.2

Phase A: Warm water until 65°C.
Phase B: Melt ingredients together at 65°C, then add to Phase A.
Use a dispersion mix to emulsify.
Slowly cool down the emulsion until 40°C.
Phase C: Introduce ingredients in given order.
Control and adjust pH, if necessary, with citric acid.

Controls: pH = 2.8 | Viscosity (Brookfield, LV2 V3, at 20°C) = 37800cP

Sodium Benzoate & Potassium Sorbate have a strong negative impact on viscosity and are consequently not recommended for hair masks

Formulation Guidelines

Making a 2-in-1 Shampoo



Phase	Ingredients	w/w %
A	Aqua/Water	Up to 100
	Hydroxyethylcellulose	1.0
	AMPHOSOL® CG-UN Cocamidopropyl Betaine	20.0
B	STEPANQUAT Soleil – Dioleoylethyl Hydroxyethylmonium Methosulfate & Sunflower Seed Oil Glycerides	1.5
	STEPAN-MILD® GCC – Glyceryl Caprylate/Caprate	1.0
	Butyrospermum Parkii (Shea) butter	0.4
	Cetyl Alcohol	3.0
	Tocopherol	0.1
C	STEPANOL WA-EXTRA-E – Sodium Lauryl Sulfate	30.0
	Citric Acid	q.s pH = 4
D	Euxyl K712 – Sodium Benzoate & Potassium Sorbate (Schülke & Mayr SA)	0.9
	Fragrance – Kiwi (Mane & Fils SA)	0.3
	Citric Acid	q.s

Phase A: Warm water until 60°C, add hydroxyethylcellulose (HEC) and mix until completely dispersed before adding Cocamidopropyl Betaine (CAPB).
Phase B: Melt ingredients together at 65°C, then add to Phase A and mix until homogeneous.
Phase C: Adjust SLS pH down to 4 before adding to Phase A+B. Start to cool down.
Phase D: Add preservative then adjust viscosity if necessary with sodium chloride, then add fragrance and mix until homogeneous. Adjust pH to 4.5 if necessary.

Controls: pH = 4.5 | Viscosity (Brookfield, LV3 V12, at 20°C) = 28 500cP

NB: HEC recommended polymer due to compatibility with cationics and ability to create a stabilizing network.
Anionic/CAPB ratio and surfactants' actives level are key for thickening and stability.
Try to reduce as much as possible the incorporation of air during the process because it remains trapped.

Explore, Enjoy, Be Creative!

Bear to Beard – Conditioning Serum



Phase	Ingredients	w/w %
A	Olea Europea (Olive) Oil	17.2
	Corylus Avellana (Hazelnut) Seed Oil	7.4
	Balanites Roxburghii (Date) Seed Oil	7.4
B	STEPAN-MILD GCC – Glyceryl Caprylate/Caprate	7.4
	STEPANQUAT Soleil – Dioleoylethyl Hydroxyethylmonium Methosulfate & Sunflower Seed Oil Glycerides	7.4
	STEPAN-MILD L3-G/MB – Lauryl Lactyl Lactate	5.0
	Oleyl Alcohol	24.6
	NEOBEE M-20 – Propylene Glycol Di (Caprylate/Caprate)	22.1
	Fragrance – Dandy Moderne (Mane & Fils SA)	1.5
	Tocopherol	0.1

Operate at room temperature and gentle mixing
Phase A: Mix the vegetable oils together until homogeneous
Phase B: Add each ingredient to phase A in given order

Controls: Appearance = clear, green olive to brown oil

Alternatives to NEOBEE M-20: NEOBEE M-5 COSMETIC (Caprylic/Capric Triglycerides), Isopropyl Myristate, Isopropyl Palmitate, Dicaprilyl Carbonate, Propylene Glycol.

Explore, Enjoy, Be Creative!

Abracada'Bar – Solid Conditioner



Phase	Ingredients	w/w %
A	Butyrospermum Parkii (Shea) Butter	3.5
	Ascorbyl Palmitate (and) Tocopherol – Sunflower Seed Wax (Kahl GmbH & Co. K.G.)	2.0
	Stearyl Alcohol	23.0
	STEPAN-MILD GCC – Glyceryl Caprylate/Caprate	7.4
	NEOBEE® M-5 COSMETIC – Caprylic/Capric Triglycerides	4.0
	Helianthus Annuus (Sunflower) Seed Oil	6.0
B	STEPANQUAT Soleil – Dioleoylethyl Hydroxyethylmonium Methosulfate & Sunflower Seed Oil Glycerides	14.0
	Sucrose Stearate – Sisterna SP70-C (Sisterna B.V.)	8.0
	Oryza Sativa (Rice) Starch – NativaCare 8600 (Ingredion)	25.0
	Decyl Glucoside – Oramix NS 10 (SEPPIC SA)	8.0
C	Lactic Acid	q.s
	Fragrance – Lime Coco (Technico Flor SA)	0.1

Phase A: Add all ingredients in given order then melt at 80°C
Phase B: Add each ingredient to phase A in given order and increase mixing speed to ensure proper dispersion of sucrose stearate and starch. Decyl glucoside will help dispersing. Cool down to 60°C after its introduction in order to avoid excess evaporation of its water.
Phase C: Control and adjust pH to 3.5-4 with lactic acid. Add fragrance at 60°C and pour into silicone molds. Once completely cooled down and solidified, remove the bars from the mold
NB: a dye can also be added for more colorful bars

Controls: Appearance = white, smooth solid

STEPAN-MILD® GCC SERIES

Hair Enhancer Conditioner

STEPAN-MILD GCC/MB*

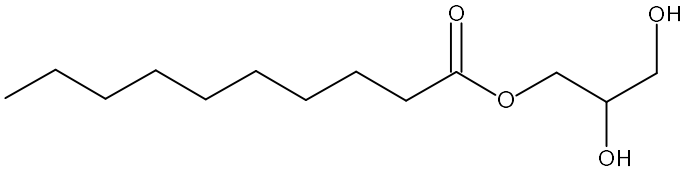
INCI Name	Glyceryl Caprylate/Caprate
Appearance	Paste
Active	100%
Preservative	None

Foam Booster
Skin Feel Enhancer
Conditioning Properties



Verified by ECOCERT
COSMOS APPROVED

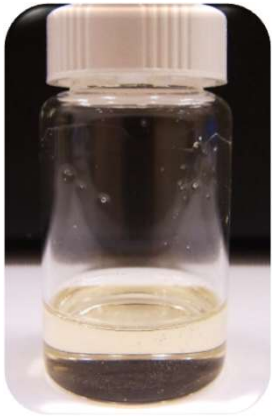
100% Bio-based
Readily biodegradable
No preservative
Cosmos approved



STEPAN-MILD GCC-FL/MB*

INCI Name	Glyceryl Caprylate/Caprate & Glycerin & Propanediol
Appearance	Liquid Viscous
Active	62%
Preservative	None

Structured Liquids
Easily thickens sulfate-free systems
Easy handling and formulating
Achieve water-white formulas



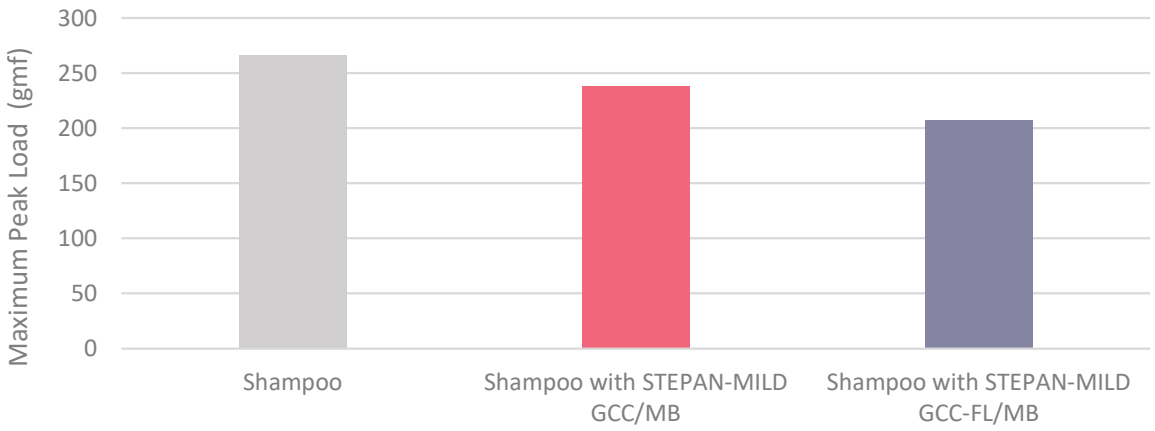
*MB designation represents RSPO Mass Balance-certified ingredients

Tangles and Shine

STEPAN-MILD GCC-FL/MB boosts detangling and shine on 2A Hair Type

Decreased Tangles

Dry Combing via Dia-Stron



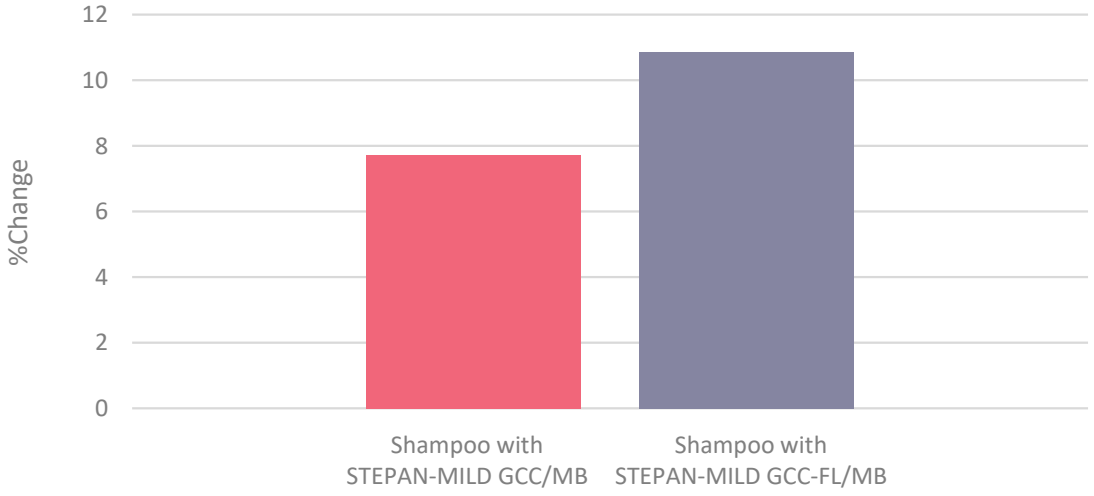
Shampoo
with cationic derivative of Hydroxypropyl guar

Versus

Shampoo
with cationic derivative of Hydroxypropyl guar
+
1% active
STEPAN-MILD GCC SERIES

Increased Shine

Shine via SAMBA
1% Active STEPAN-MILD GCC Series

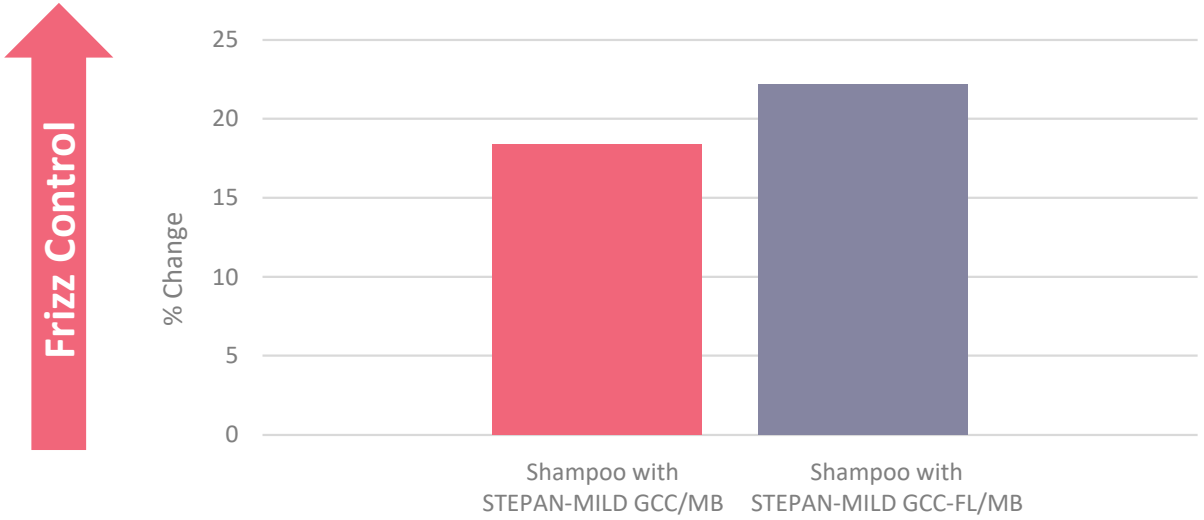




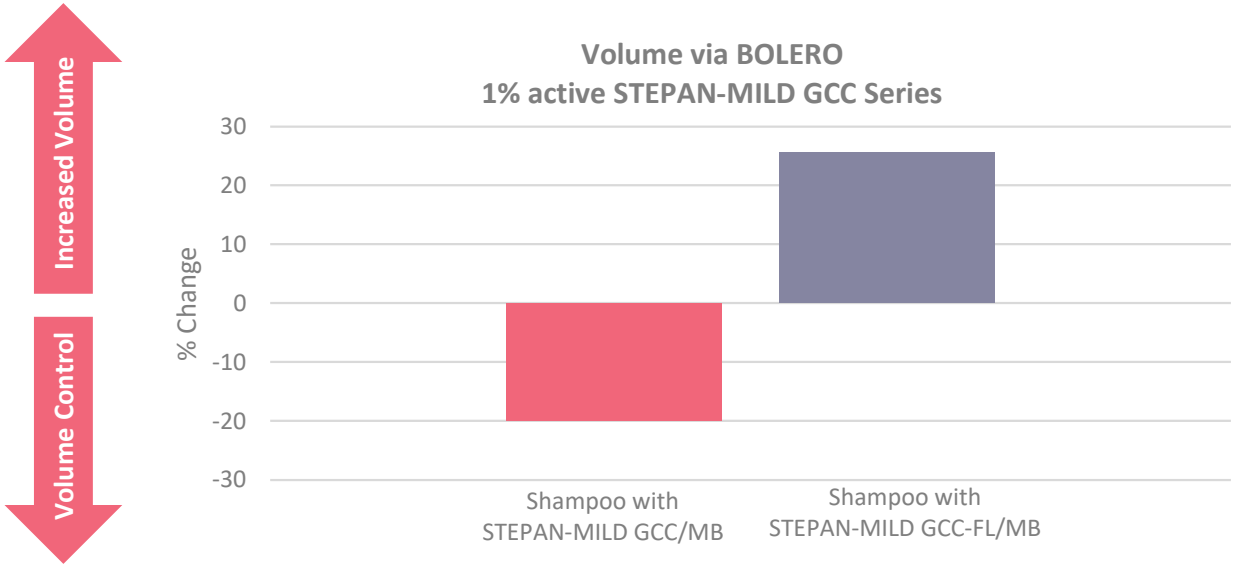
Flyaway and Volume

STEPAN-MILD GCC-FL/MB reduce flyaway and frizz and increase volume on 2A Hair Type

Frizz Reduction via BOLERO
1% Active STEPAN-MILD GCC Series



Volume via BOLERO
1% active STEPAN-MILD GCC Series

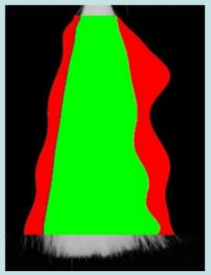


BOSSA NOVA BOLERO
Analyzed Images of Hair tresses

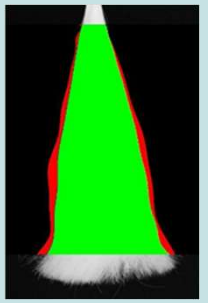
Type 2A

Flyaway Volume

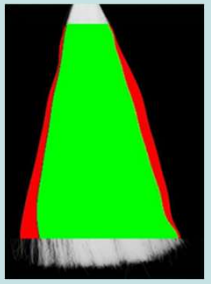
Shampoo



STEPAN-MILD GCC/MB



STEPAN MILD GCC-FL/MB



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